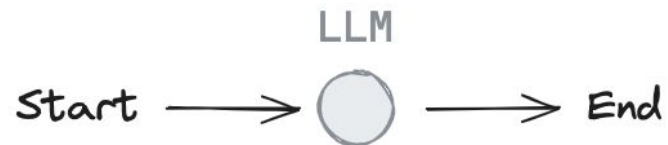


LangChain Academy

Introduction to  LangGraph

A solitary language model is fairly limited...



... e.g., access to tools, external context, multi-step workflows.

So, many LLM applications use a control flow ...



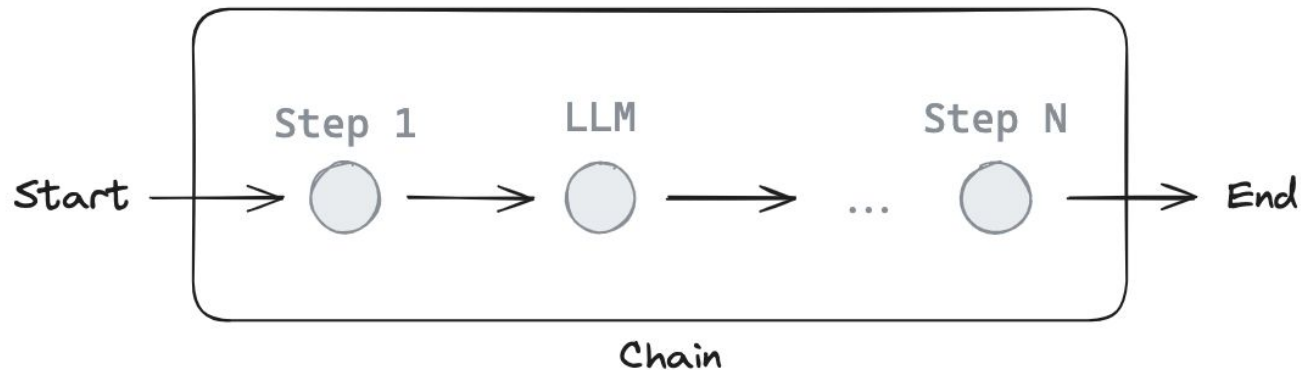
... with steps pre / post-LLM call (tool calls, retrieval, etc).

Reference:

https://en.wikipedia.org/wiki/Control_flow

<https://github.com/langchain-ai/rag-from-scratch>

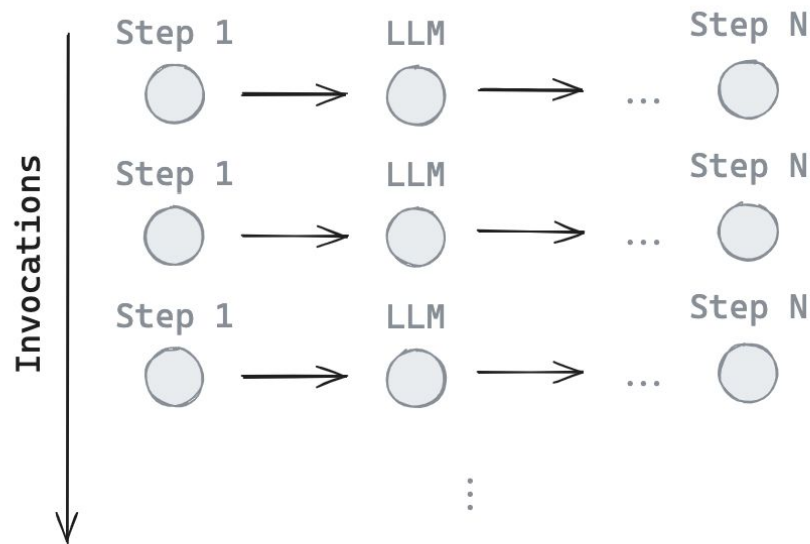
This control flow forms a “chain”



Reference:

https://en.wikipedia.org/wiki/Control_flow
<https://blog.langchain.dev/what-is-an-agent/>

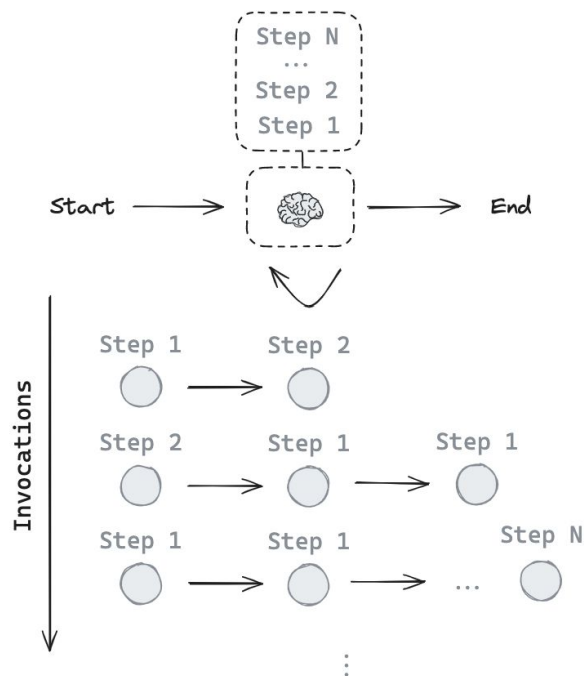
Chains are reliable! Same control flow every time.



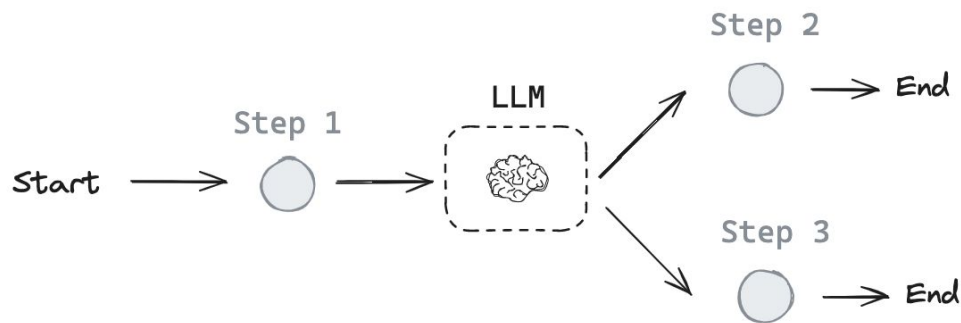
Reference:

https://en.wikipedia.org/wiki/Control_flow
<https://blog.langchain.dev/what-is-an-agent/>

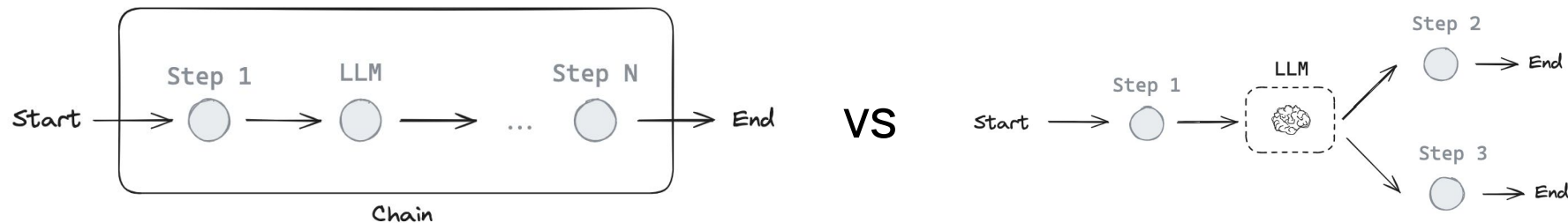
But, we want LLM systems that can pick their own control flow!



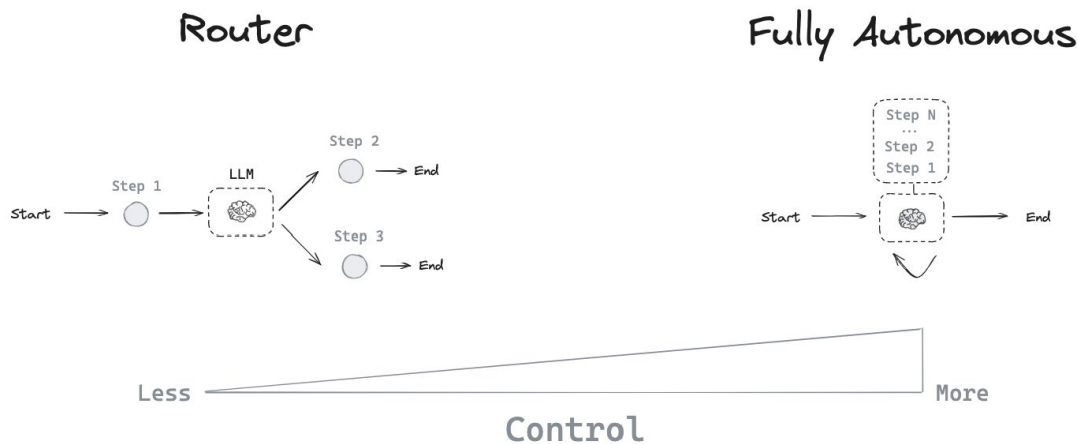
Agent $\sim$ control flow defined by an LLM



Fixed vs LLM-defined control flow



Many kinds of agents!

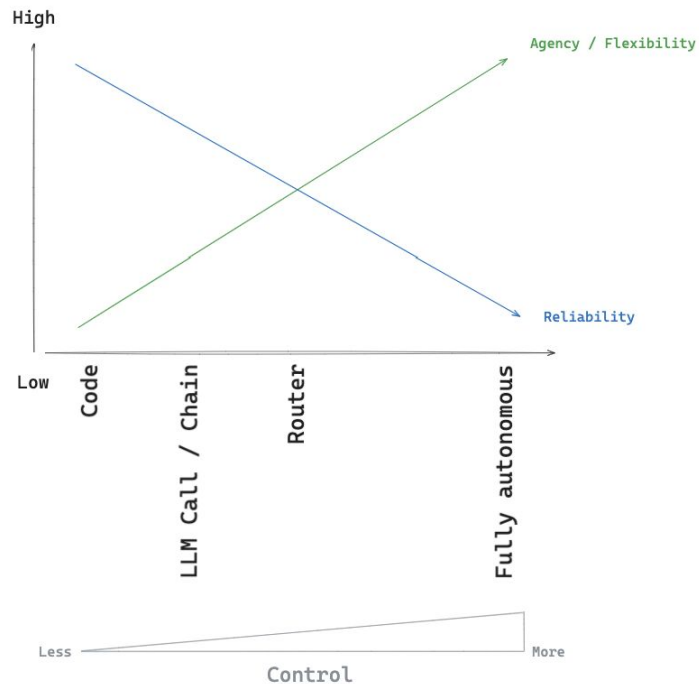


Reference:

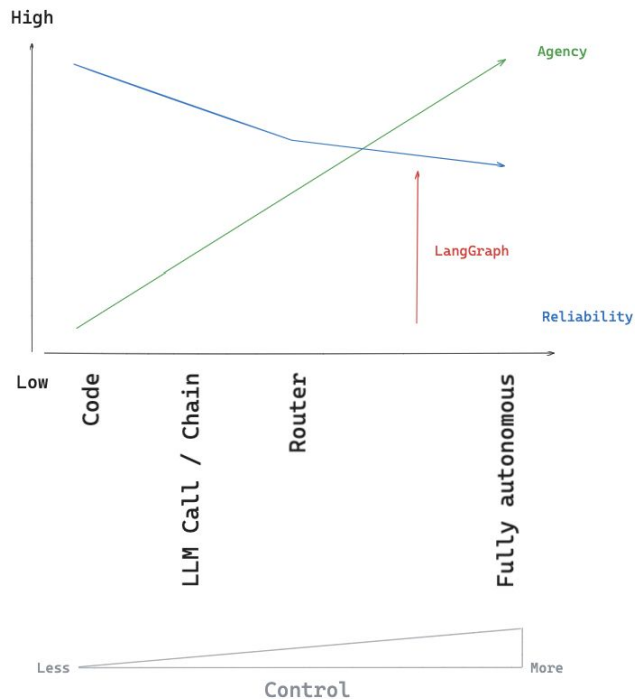
<https://blog.langchain.dev/what-is-a-cognitive-architecture/>

<https://x.com/AndrewYNg/status/1801295202788983136>

But, practical challenges.



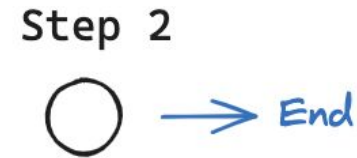
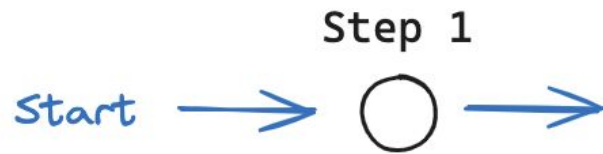
LangGraph helps you bend the reliability curve



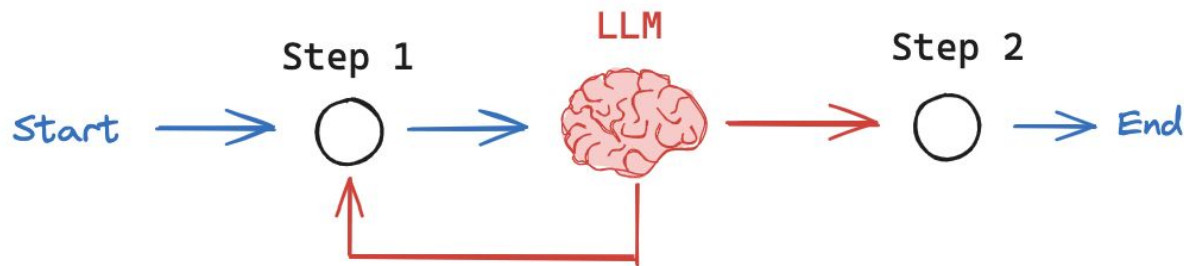


Balance reliability with control

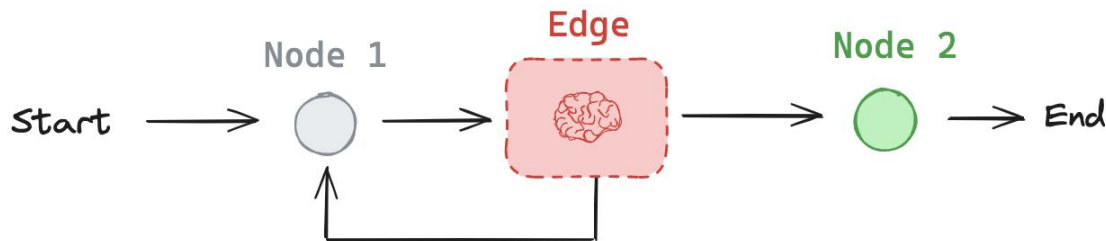
Intuition: Let developer set parts of control flow (reliable)



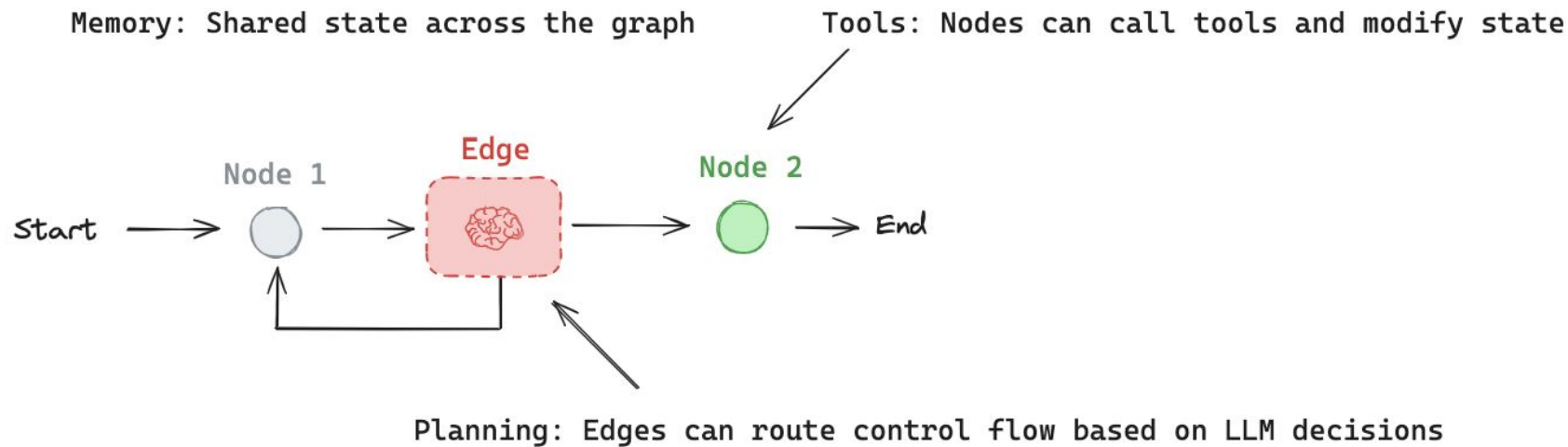
Intuition: Inject LLM to make it an agent (control)



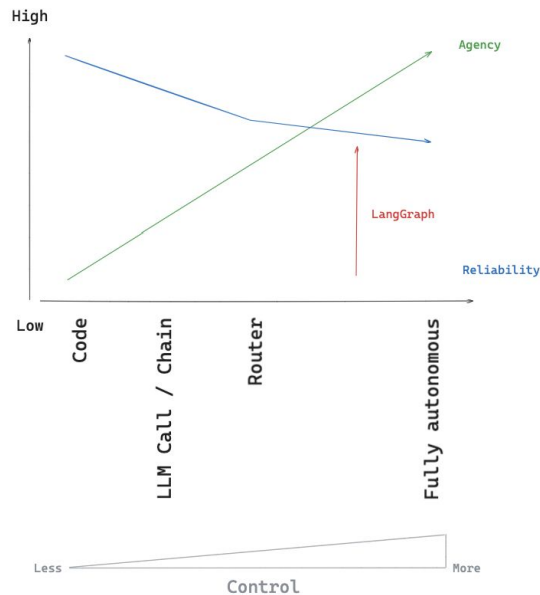
LangGraph Express custom control flows as graphs



1 Introduction and Basics: Motivation



1 Introduction and Basics: Motivation



Module	What you will learn
1: Foundations	LangGraph foundations Chains Routers General autonomous agents
2: Memory	Agents that remember things
3: Human-In-The-Loop	Agents with human oversight
4. Customization	Customized, useful agents